

Experiment 01: Input & Output Manual

Design a Registration Form using HTML5 and CSS3

Student & Laboratory Credentials

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- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Design a Registration Form using HTML5 and CSS3 on Selected Problem Statement

1. Experiment Aim

To design, implement, and validate a responsive, accessible, and secure Patient Health Profile & HIPAA Insurance Registration Form using semantic HTML5 elements and modern CSS3 styling with real-time client-side constraint validation.

2. Input Specification

2.1 File System Input

The following source files constitute the application input:

File Name	Language / Format	Role & Implementation Responsibility
index.html	HTML5	Semantic structure (<code><form></code> , <code><fieldset></code> , <code><legend></code> , <code><input></code> , <code><select></code> , <code><button></code>) with validation attributes (<code>required</code> , <code>pattern</code> , <code>minlength</code> , <code>maxlength</code>).
style.css	CSS3	Responsive CSS Grid & Flexbox, clinical blue color variables (<code>:root</code>), <code>:focus</code> glow states, and <code>:valid</code> / <code>:invalid</code> pseudo-class indicators.
script.js	JavaScript (ES6)	Client-side password matching logic, submission interception, and dynamic patient summary confirmation modal rendering.

2.2 Execution Command Input

To launch and test the application, execute any of the following commands:

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-01-Registration-Form

# Option A: Open directly in Google Chrome / Chromium browser
chromium-browser index.html

# Option B: Run via lightweight HTTP server
npx serve .
# Or via Python standard library:
python3 -m http.server 8080
```

2.3 Form Fields & Constraint Validation Rules

FIELD NAME	INPUT TYPE	VALIDATION CONSTRAINT / REGEX PATTERN	ERROR BEHAVIOR
Full Name	text	pattern="^[a-zA-Z\s]+\$", minlength="3"	Blocks numbers/symbols
Date of Birth	date	required, max="2026-01-01"	Prevents future dates
Gender	radio	required (Male / Female / Other)	Selection mandatory
Blood Group	select	required (A+, A-, B+, B-, O+, O-, AB+, AB-)	Must choose from list
Phone Number	tel	pattern="[6-9][0-9]{9}", maxlength="10"	Enforces 10 digits 6-9
Email Address	email	required, standard RFC 5322 syntax	Requires '@' & domain
Address	textarea	required, minlength="10"	Minimum description
Chronic Illnesses	checkbox	optional (Diabetes, Hypertension, etc.)	Multiple selection
Insurance Number	text	pattern="^[A-Z0-9-]{6,16}\$"	Alphanumeric & dashes
Account Password	password	(?=.*\d)(?=.*[a-z])(?=.*[A-Z])(?=.*[\W]).{8,}	Complex pass rule
Confirm Password	password	Value must strictly match Password field	Triggers inline alert
Medical ID Document	file	accept=".pdf,.png,.jpg,.jpeg"	Restricts file format
Terms & HIPAA Consent	checkbox	required (boolean true)	Blocks submit if false

2.4 Sample Input Test Vectors

Test Case A: Valid Full Enrollment Submission

```
{
  "fullName": "Asmit Jogdand",
  "dob": "2004-08-15",
  "gender": "Male",
  "bloodGroup": "O+",
  "phone": "9876543210",
  "email": "asmit.jogdand@healthpulse.org",
  "emergencyContact": "9820112233",
  "address": "Flat 402, Seawoods Grand Central, Sector 40, Nerul, Navi Mumbai - 400706",
  "chronicConditions": ["Hypertension", "Asthma"],
  "allergies": "Penicillin",
  "insuranceProvider": "Star Health Allied Insurance",
  "insurancePolicyNo": "POL-992144-HP",
  "accountPassword": "HealthPulse@2026!",
  "confirmPassword": "HealthPulse@2026!",
  "documentAttachment": "asmit_aadhaar_card.pdf",
  "hipaaConsent": true
}
```

Test Case B: Boundary & Erroneous Input Vector (Validation Testing)

```
{
  "fullName": "Asmit123", // Invalid: Contains numeric digits
  "dob": "2030-05-20", // Invalid: Future date exceeding max bound
  "phone": "555123", // Invalid: Only 6 digits and doesn't start with 6-9
  "email": "asmitjogdand.healthpulse", // Invalid: Missing '@' symbol and TLD domain
  "accountPassword": "pass", // Invalid: Less than 8 characters, lacks uppercase/symbol
  "confirmPassword": "mismatchPassword", // Invalid: Mismatched password
  "hipaaConsent": false // Invalid: Checkbox unchecked
}
```

3. Output Specification

3.1 Graphical User Interface Output Representation

When loaded in the browser, the application displays a modern healthcare registration interface:

```

+-----+
| [HealthPulse Portal] Skill Based Lab - Advanced Web Technology (Sem VI) Student: Asmit (25CE1051)|
+-----+
|
|           ▢ HEALTHPULSE DIGITAL CLINIC
|       New Patient Medical Profile & Enrollment Portal
|
|
|  +-- 1. Patient Demographics -----+
|  | Full Legal Name: [ Asmit Jogdand           ] Date of Birth: [ 15/08/2004   ] |
|  | Gender: (o) Male   ( ) Female   ( ) Other   Blood Group: [ 0+             v] |
|  | Primary Cellular: [ +91 9876543210         ] Email Address: [ asmit@hp.org   ] |
|  | Residential Address: [ Flat 402, Seawoods Grand Central, Nerul, Navi Mumbai ] |
|  +-----+
|
|  +-- 2. Medical History & Insurance Credentials -----+
|  | Pre-Existing Chronic Conditions:
|  | [X] Hypertension [ ] Diabetes Mellitus [X] Asthma / Respiratory [ ] None Known
|  | Known Allergies: [ Penicillin, Sulfa Drugs                               ]
|  | Insurance Provider: [ Star Health Allied Insurance ] Policy ID: [ POL-992144-HP ]
|  | Upload ID Proof (PDF/JPG): [ Choose File ] asmit_aadhaar_card.pdf
|  +-----+
|
|  +-- 3. Security Credentials & Consent -----+
|  | Create Password: [ ..... ] [👁 Show] (Password Entropy: Strong ✓)
|  | Confirm Password: [ ..... ] (Passwords Match ✓)
|  | [X] I declare that all medical info is accurate and accept HIPAA data policies.
|  +-----+
|
|           [ 🔄 Reset Form ]           [ ▢ Submit Patient Registration ]
+-----+

```

3.2 Submission Success Modal Dialog Output

Upon clicking "**Submit Patient Registration**" with valid input data, `script.js` intercepts the submit event, validates all constraints, builds a modal card, and displays the confirmation summary:

```

+-----+
|           ▢ PATIENT ENROLLMENT CONFIRMED
+-----+
| HealthPulse Record Identifier: HP-2026-992144
| Registration Status:           ACTIVE / VERIFIED
| Timestamp:                     2026-10-01 10:15:30 IST
|
| Patient Summary:
| -----
| • Full Legal Name:   Asmit Jogdand
| • Date of Birth:     15 August 2004 (Age: 22)
| • Blood Group:       0 Positive (O+)
| • Contact:           +91-9876543210 | asmit.jogdand@healthpulse.org
| • Insurance Policy:  Star Health Allied Insurance (POL-992144-HP)
| • Medical Alerts:    Hypertension, Asthma | Allergy: Penicillin
| • Document Attached: asmit_aadhaar_card.pdf (Verified)
|
| [ Close & Download Patient Summary Slip ]
+-----+

```

3.3 Test Case Execution Results

Test Scenario	Test Input Injected	Expected System Behavior	Actual Observed Output	Status
Empty Submission	All fields blank	Browser Constraint API blocks submission	Native HTML5 tooltip: "Please fill out this field" appears on Name	PASS
Numeric Name	12345	Reject input matching pattern <code>^[a-zA-Z\s]+\$</code>	Input border turns red, tooltip: "Please match the requested format"	PASS
Future DOB	2030-01-01	Reject date exceeding <code>max="2026-01-01"</code>	Date picker disallows selection beyond maximum allowed date	PASS

Test Scenario	Test Input Injected	Expected System Behavior	Actual Observed Output	Status
Invalid Cellular	98201 (5 digits)	Reject length less than 10 digits	Browser blocks submit: "Please match requested pattern"	PASS
Malformed Email	asmit.healthpulse	Enforce RFC email format	Native tooltip: "Please include an '@' in the email address"	PASS
Weak Password	password123	Reject missing uppercase and symbol	Tooltip flags pattern requirement	PASS
Password Mismatch	Health@2026! vs Health@2026?	Client JS checks val1 === val2	Inline red alert text: "Passwords do not match!"	PASS
Valid Full Data	Asmit Jogdand complete profile	Form passes validation and triggers modal	Confirmation modal pops up with complete patient record	PASS

4. Conclusion & Verification

Experiment 01 was executed successfully. Semantic HTML5 elements ensured accessible document structure, CSS3 responsive grid rules adapted fluidly across display viewports, and both HTML5 Constraint Validation API and JavaScript DOM listeners verified all patient registration parameters before submission.